SEC 2 curves: identity and structural traits

2026-10-12 · primary-standard crosscheck · 33 named curves

The 2000 edition lists 33 curves. The 2010 edition retains 20 of them; 13 are historical entries. Every field, coefficient, uncompressed base point, subgroup order, and cofactor in both editions matched the exact crypto registry record.

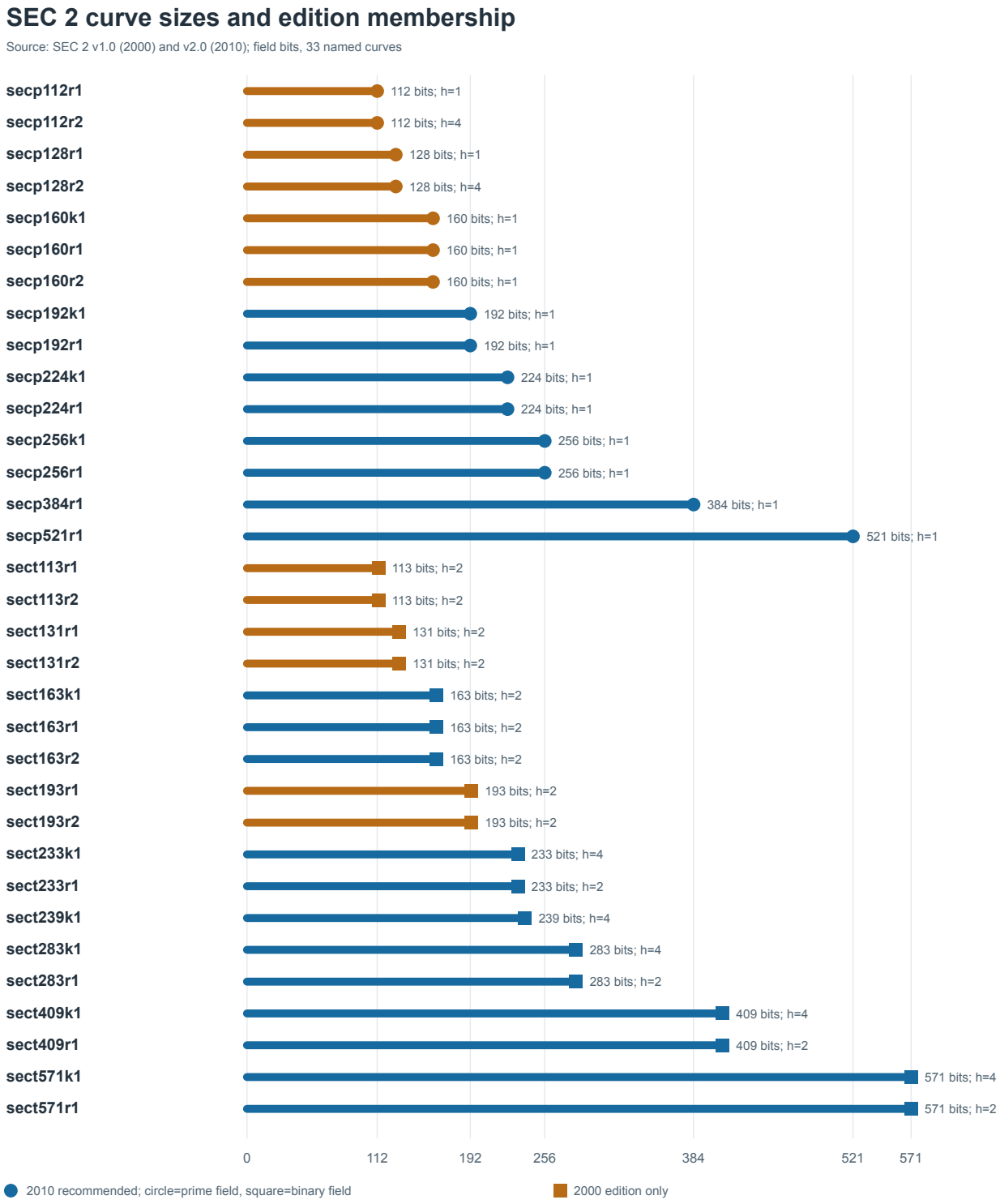


Figure 1: Field size and edition membership. Circle: prime field; square: binary field. Blue: listed in 2010. Amber: 2000 only.

Curve index

Full exact identities, seeds, small-prime splitting, conductors, embedding and twist records, cover and transfer screening, and all status fields are in `inventory.json`. F in the last column means the SEC 1 advisory bound fails; ? means the bounded n±1 check was inconclusive.

Curve	Edition	Field	h	CM status	Twist	n±1
secp112r1	2000 only	Fp/112	1	probable	proved	F/?
secp112r2	2000 only	Fp/112	4	probable	proved	F/?
secp128r1	2000 only	Fp/128	1	bounded	proved	F/F
secp128r2	2000 only	Fp/128	4	probable	probable	F/F
secp160k1	2000 only	Fp/160	1	−3 (proved)	probable	F/F
secp160r1	2000 only	Fp/160	1	probable	probable	?/F
secp160r2	2000 only	Fp/160	1	probable	probable	F/F
secp192k1	2010	Fp/192	1	−3 (proved)	probable	F/F
secp192r1	2010	Fp/192	1	probable	unknown	F/?
secp224k1	2010	Fp/224	1	−3 (proved)	unknown	?/F
secp224r1	2010	Fp/224	1	probable	unknown	F/?
secp256k1	2010	Fp/256	1	−3 (probable)	probable	F/F
secp256r1	2010	Fp/256	1	bounded	probable	F/F
secp384r1	2010	Fp/384	1	bounded	probable	?/F
secp521r1	2010	Fp/521	1	bounded	probable	?/?
sect113r1	2000 only	F2m/113	2	proved	proved	F/F
sect113r2	2000 only	F2m/113	2	probable	proved	F/F
sect131r1	2000 only	F2m/131	2	probable	probable	F/?
sect131r2	2000 only	F2m/131	2	probable	probable	?/F
sect163k1	2010	F2m/163	2	−7 (proved)	unknown	F/F
sect163r1	2010	F2m/163	2	bounded	unknown	F/F
sect163r2	2010	F2m/163	2	bounded	probable	F/?
sect193r1	2000 only	F2m/193	2	probable	unknown	F/F
sect193r2	2000 only	F2m/193	2	probable	unknown	?/?
sect233k1	2010	F2m/233	4	−7 (proved)	unknown	F/F
sect233r1	2010	F2m/233	2	bounded	unknown	F/?
sect239k1	2010	F2m/239	4	−7 (proved)	unknown	F/?
sect283k1	2010	F2m/283	4	−7 (probable)	probable	F/?
sect283r1	2010	F2m/283	2	bounded	unknown	F/?
sect409k1	2010	F2m/409	4	−7 (proved)	unknown	F/F
sect409r1	2010	F2m/409	2	bounded	unknown	?/?
sect571k1	2010	F2m/571	4	−7 (probable)	unknown	F/F
sect571r1	2010	F2m/571	2	bounded	unknown	?/?

Sources, method, and status

SEC 2 v1.0 (2000): `https://www.secg.org/SEC2-Ver-1.0.pdf` SHA-256 `d1b16728ad83888fd656d16b99dc71bcd5541d42d848ffd0de7c62c19010d8c3`. SEC 2 v2.0 (2010): `https://www.secg.org/sec2-v2.pdf` SHA-256 `87b8f3703364ed5b21ba8582e411cc0cbf477bcaa3f4f45e0d6580d1c00d9952`.

The two PDFs supply 33 and 20 exact named parameter blocks. `build.py` compares field prime or reduction polynomial, coefficients, uncompressed base point, prime subgroup order, and cofactor with the registry. All 53 blocks match. Seeds are transcribed but the ANSI seed-to-curve derivation is not independently replayed.

Existing `curve_traits` records give subgroup, CM/conductor, endomorphism bounds, small-prime splitting, subfield, embedding, and twist data. Cover and Jacobian-transfer screening rows match all 33 exact identities. Every cover has subgroup transport `not_tested`; the transfer formulas are screening models.

The bounded n±1 scan certifies failure of the later SEC 1 19/20-factor advisory for 25 n−1 values and 19 n+1 values. Among curves retained in 2010, 15 n−1 and 10 n+1 values also fail, so failure of this advisory condition alone cannot account for the 193-bit omission. The other cases remain unresolved. The trait and transfer files have older whole-registry pins, but all 33 selected rows match current exact identities and order/trace.

Evidence boundaries

This is a primary-parameter check and a structured view of existing catalog traits. A `probable` or `bounded` trait is not promoted to proved. The n±1 scan uses trial division through 100,000 and proves failure of the SEC 1 advisory threshold only when the residual is already below it. Covers and transfer screening do not verify subgroup transport or measure attack cost. Full status-bearing records are in `inventory.json`; source PDF copies are in `sources/`.